MILITARY SPECIFICATION

SWITCHBOARD, TRANSMITTER CONTROL, SB-3916/BRT

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a transmitter control switchboard, SB-3916/BRT, hereinafter referred to as the switchboard, used to interconnect radio transmitters to various local and remote sources on board submarines. Signal lines to be switched consist of alternating current (AC), direct current (DC), and audio.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-5015	Connectors, Electrical, Circular Threaded, AN Type, General Specification For
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-E-16400	Electronic Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-C-26482	Connectors, Electrical, (Circular, Miniature, Quick Disconnect, Environment Resisting), Receptacles And Plugs, General Specification For
MIL-C-28777/1	Cable Assembly, Electronic Test Equipment, (3 Wires, 3 Prongs, 125 Volts Molded) Grounding Plug Connector, And MIL-C-5015 Connector With Cable Clamp, 60 Hertz

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-189	Racks, Electrical Equipment, 19-Inch And Associated Panels

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1378	Requirements For Employing Standard Electronic Modules
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70-1978	National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The switchboard shall be in accordance with MIL-E-16400, except as hereinafter specified.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Description. The switchboard shall interconnect eight remote sources and eight radio transmitters by means of eight, 8-position, solenoid-driven rotary switches. The rotary switches shall be controlled electrically via thumbwheel switches on the front panel of the enclosure.

3.2.1 Crosspoint diagram. The crosspoint switching of the switchboard shall be in accordance with the crosspoint diagram shown in FIGURE 1.

3.2.2 Interface definition. The switchboard crosspoints shall be capable of carrying the following types of signals:

a. 600-ohm balanced audio (0 kilohertz (kHz) to 5 kHz)

b. ±28 volts direct current (VDC)

c. ±12 VDC

d. 115 volts alternating current (VAC), 60 hertz (Hz) or 400 Hz

e. 600-ohm unbalanced audio (0 kHz to 20 kHz)

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TRANSMITTER	SOURCE								DISPLAY READOUT	OUTPUT JACK	SWITCH POSITION
	C-1138 Remote hand set	KY-75	C-1004	CV-2460	SPARE	KY-8	USQ-76 USB 1/	USQ-76 LSB 2/			
	SWITCH AND INPUT-JACK NUMBER										
	1	2	3	4	5	6	7	8			
URT-23 #1 USB	X	X	X	X	X		X		URT-23 1 USB	J9	1
URT-23 #1 LSB	X	X	X		X			X	URT-23 1 LSB	J10	2
URT-23 #2 USB	X	X	X	X	X		X		URT-23 2 USB	J11	3
URT-23 #2 LSB	X	X	X		X			X	URT-23 2 LSB	J12	4
WSC-3 #1			X	X	X	X	X	X	WSC-3 #1	J13	5
WSC-3 #2			X	X	X	X	X	X	WSC-3 #2	J14	6
AN/BSQ-5		X		X	X	X			BSQ-5	J15	7
SPARE #1	X	X	X	X	X	X	X	X	SPARE #1	J16	8, 9, 0

NOTE: X indicates valid connections

1/ Upper sideband

2/ Lower sideband

FIGURE 1. Crosspoint diagram.

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3.2.2.1 Insertion loss. The insertion loss of the switchboard shall be less than 0.1 decibel (dB) from 20 Hz to 20 kHz.

3.2.2.2 Characteristic impedance. The nominal characteristic impedance of each input and output terminal shall be 600 ohms.

3.2.2.3 Crosstalk isolation. Crosstalk isolation between input terminals shall be 60 dB minimum.

3.3 Characteristics.

3.3.1 Transmitter selection. Desired selections shall be chosen by means of thumbwheel switches located on the front panel. Each thumbwheel switch shall represent a source, and positions on the rotary switch shall represent the transmitters.

3.3.2 Displays. When an input/output (I/O) connection has been selected, the name of the selected transmitter shall be displayed on the screen of a rear projection device. The signal activating the display shall originate at the rotary switch.

3.3.2.1 Display nomenclature. The nomenclature displayed on the rear projection device shall be as shown under DISPLAY READOUT column in FIGURE 1.

3.3.3 Controlled switches. The transmitter controlled switches shall be multideck, power-driven, rotary switches whose positions are selected by eight thumbwheel switches.

3.3.3.1 Rotary switch capacity. Each solenoid-driven rotary switch shall have an 8-position, 16-pole capacity.

3.3.3.2 Rotary switch function by section. Rotary switches shall be wired as shown in TABLE I.

TABLE I. Rotary switch wiring.

Section		Function	
Wafer 2,	A	Power-on indication	
	B	Start bus	
	C	Stop bus	
Wafer 3,	A	Power-bus ground	
	B	Key ring	
	C	Key tip	
Wafer 4,	A	+12 V (or ground)[1/]	
	B	Ground (or -12 V)[1/]	
	C	600-ohm audio	
Wafer 5,	A	Balanced	
	B	12-V carrier on	
	C	Alert	
Wafer 6,	A	Wideband control-KY-8 only	
	B	Data keyline	AN/WSC-3 only
	C	Data keyline return	AN/WSC-3 only

1/ Wafer 4, section A, shall be +12 V with respect to Wafer 4, section B.

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3.3.3.3 Switch identity. Each rotary switch shall be identified on a one-to-one basis with a thumbwheel selection switch.

3.3.3.4 Manual override. Manual override shall be provided for each rotary switch in case of power failure.

3.3.3.5 Power. Power to operate the rotary switches shall be controlled independent of the power to operate the displays.

3.3.4 Alert circuit. If more than one source is connected to the same transmitter, the alert circuit shall be activated.

3.3.4.1 Capability. The alert circuit shall have a full 8 by 8 capability.

3.3.4.2 Applicable crosspoints. Only those crosspoints shown in FIGURE 1 shall be connected to the alert circuitry.

3.3.5 Indicators and controls. The switchboard shall be provided with the indicators and controls specified in 3.3.5.1 through 3.3.5.2. The design of controls, indicators, and their arrangement on operator and maintainer panels, shall conform to the criteria of MIL-STD-1472 and shall be in accordance with the Controls, indicators and panel layout paragraph of MIL-E-16400.

3.3.5.1 Power.

3.3.5.1.1 Primary switch. An AC primary power indicator switch shall be provided to connect operating power to the switchboard. The switch shall break all power conductors of the power circuits.

3.3.5.1.2 Primary power indicator. When the primary power switch is turned on, a green indicator mounted near the switch shall be illuminated. The indicator lamp shall be connected to the load side of the switch and across the input power conductors.

3.3.5.1.3 Solenoid control. A front panel switch shall be provided to connect power to the solenoid circuitry of the rotary switches.

3.3.5.1.4 Solenoid indicator. When the solenoid power switch is activated, a second green indicator shall be illuminated.

3.3.5.2 Alert indicator. When the alert conditions specified in 3.3.4 are violated, a red indicator shall be illuminated.

3.3.6 Operating power. Any power supply required for operation of the system shall be an integral part of the switchboard.

3.3.7 Power failure. In case of power failure to the switchboard, all I/O connections shall remain connected.

3.4 Physical characteristics.

3.4.1 Passage requirements. When uncrated, the switchboard shall be capable of passing through a doorway 50.8 centimeters (cm) (20 inches (in.)) wide by 96.52 cm (38 in.) high (reduced by rounded corners on a 25.4 cm (10 in.) radius and through a 62.50 cm (25 in.) diameter circular tube.

3.4.2 Degree of enclosure. The switchboard enclosure shall be dripproof (45 degrees) as specified in the Dripproof paragraph of MIL-STD-108.

3.4.3 Mounting. The switchboard (with enclosure) shall be capable of being mounted in a standard 48.26 cm (19 in.) relay rack, as specified in MIL-STD-189.

3.4.4 Size and weight. The overall switchboard height shall not exceed 13.1 cm (5.25 in.). The depth from front panel surface shall not exceed 50.8 cm (20 in.). The weight of the switchboard shall not exceed 22.5 kilograms (50 pounds).

3.4.5 Power source. The switchboard shall be designed to operate from a line 115 VAC ±10 percent single-phase, 60 Hz ±5 percent. Nominal steady-state power shall be on the order of 15 watts (W) with surges to approximately 100 W for very short periods while the rotary switches are being driven.

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3.4.6 Reliability.

3.4.6.1 Quantitative reliability requirements. The upper test mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) shall be 2200 hours. The lower test MTBF (θ_1 as defined by MIL-STD-781) shall be 1100 hours.

3.4.6.2 Component reliability. The reliability of the major components of the switchboard shall be as specified in 3.4.6.2.1 and 3.4.6.2.2.

3.4.6.2.1 Switches. All switches and relays shall have a minimum life of 100,000 operations.

3.4.6.2.2 Displays. All displays shall have a minimum life of 25,000 hours.

3.4.7 Maintainability.

3.4.7.1 Quantitative corrective maintenance. The switchboard shall have a mean-corrective-maintenance-time (M_{ct}), not exceeding 0.5 hour and a maximum-corrective-maintenance-time (M_{maxct}), 95th percentile, not exceeding 1.5 hours when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, printed circuit (PC) boards) and chassis-mounted parts (electronic, electrical, electromechanical and mechanical parts). The corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions. The switchboard is considered to be in a corrective maintenance mode whenever a failure or malfunction occurs, including a malfunction that allows degraded operation.

3.4.7.2 Subassembly (module) maintainability. Repairable modules shall have a M_{ct} of 2.0 hours and a M_{maxct} of 4.0 hours, 95th percentile, at the organizational, intermediate, or depot level maintenance, as appropriate. M_{ct} for module repair includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all maintenance tasks. For purposes of maintainability, a module is considered the lowest replaceable subassembly (PC card) for which repair is accomplished by the replacement of piece parts.

3.4.7.3 Modular construction. Predeveloped modules from the Standard Electronic Modules (SEM) program module listing may be used. Prior approval from the contracting officer shall be obtained, before designing a SEM in accordance with MIL-STD-1378, for use in the switchboard.

3.4.8 Human engineering. The switchboard shall conform to the human engineering operational and maintenance design criteria of MIL-STD-1472 and the requirements specified herein.

3.5 Environmental conditions. The switchboard shall be designed to conform to the environmental requirements specified in 3.5.1.

3.5.1 Temperature. The switchboard shall operate under temperature ranges specified for Range 4 equipment in accordance with MIL-E-16400.

3.6 Design and construction. The design objectives of the switchboard shall be to achieve maximum reliability and minimum size while reducing mean-time-to-repair through modular construction.

3.6.1 Parts control. The parts to be incorporated in the switchboard shall be controlled in accordance with MIL-STD-965, Procedure I.

3.6.1.1 Solid-state components. Solid-state components, including microelectronic devices, shall be used to the maximum degree practicable.

3.6.1.2 Enclosure. The enclosure for the switchboard shall be of aluminum or steel. The chassis shall be slide-mounted within a rack.

3.6.1.3 Chassis access. The chassis shall extend a minimum of 10 cm (4 in.) beyond the plane of the front panel with the slide fully extended.

3.6.1.4 Blown-fuse indicators. Blown-fuse indicating fuseholders shall be used in the switchboards.

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3.6.1.4.1 Fuse location. A fuse shall be connected in series with each ungrounded conductor as determined by the National Electrical Code (NFPA 70-1978) for grounded source power and shall be located on the load side of the power switch.

3.6.1.4.2 Power cable assembly. The input power cable assembly shall be Type I in accordance with MIL-C-28777/1. Power cable connection to the switchboard shall be from the rear.

3.6.1.5 Connector location. All connectors used for inputs and outputs shall be located at the rear of the switchboard.

3.6.1.6 Connectors. The external connectors shall be in accordance with MIL-C-5015 or MIL-C-26482. Mating connectors shall be furnished.

3.6.1.7 Handles. Handles shall be provided on the front panel of the switchboard to ensure safe and easy removal of the switchboard from the equipment rack. Handles shall protrude farther than any switch, but not more than 5.00 cm (2 in.) from the front panel.

3.6.1.8 Exterior finish. The outer surfaces of the enclosure shall be protected with light-gray enamel in accordance with MIL-E-15090.

3.6.2 Electromagnetic interference (EMI) characteristics. The switchboard shall conform to the requirements of MIL-STD-461 for Class A5 equipment, except that RS03 shall be performed at 10 V per meter. If line-to-ground powerline filters are used to conform to the MIL-STD-461 requirements, the total capacitance shall not exceed 0.1 microfarad (μF) for 60 Hz switchboards or 0.02 μF for 400 Hz switchboards.

3.6.3 Front panel position card. The information shown in TABLE II shall be imprinted on a card attached to the front panel. The front panel card shall be titled Input Equipment Position Legend.

TABLE II. Input equipment position legend markings.

Position	Equipment
1	URT-23 #1 USB
2	URT-23 #1 LSB
3	URT-23 #2 USB
4	URT-23 #2 LSB
5	WSC-3 #1
6	WSC-3 #2
7	AN/BSQ-5
8, 9, 0	SPARE #1

3.6.4 Maintenance. The switchboard shall use plug-in assemblies containing subdivisions of related electronic circuit functions whenever feasible in order to reduce the required effort of maintenance and repair.

3.6.4.1 Conformal coating. To facilitate I/O equipment circuit reconfiguration, conformal coating is not required.

3.6.5 Identification of conductors. Conductors, other than printed wiring, shall be identified in accordance with MIL-E-16400.

3.6.6 Special maintenance items. Any special tools (see 6.4.2) or items, such as extender boards or cables, required for the maintenance of the switchboard, shall be provided by the contractor.

3.6.7 Workmanship screen. The switchboard shall withstand a defect detection vibration screen of random type vibration at $0.04g^2$/Hz ±3 dB from 80 Hz to 350 Hz (see 4.5.7.1). The switchboard shall withstand a temperature cycling screen to assure elimination of initial failures due to random defective components and marginal operation due to component lot variations or poor assembly processes and procedures (see 4.5.7.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

1. Production inspection (Group A) (see 4.4.1)

2. Production control inspection (Group B) (see 4.4.2)

3. Environmental inspection (Group C) (see 4.4.3)

c. Reliability testing (see 4.5.2)

d. Maintainability demonstration

e. Inspection of preparation for delivery (see 4.6)

4.3 First article inspection. Unless otherwise specified (see 6.2.1), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE III.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4 and as shown in TABLE III.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on each switchboard offered for delivery. The inspection shall comprise such examinations and tests as will provide the workmanship and reveal omissions and errors of the production process, such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls and adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests specified in TABLE III (Group A).

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.2.1. The inspection shall consist of the examinations and tests which encompass functional and performance tests throughout the entire range of operations; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; tests to determine temperature rise produced in operation and ability of the switchboards to withstand the heat; tests of efficiency; and tests of the performance with other equipment in a system. The tests shall be performed on the complete switchboard as offered for delivery. Production control inspection shall include the examinations and tests specified in TABLE III (Group B).

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TABLE III. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination						
Weight	3.4.4	4.5	X	X		
Size	3.4.1, 3.4.3, 3.4.4	4.5	X			
Parts control	3.6.1	4.5	X	X		
Finish	3.6.1.8	4.5	X	X		
Marking	3.1, 3.6.5	4.5	X	X		
Safety	3.1	4.5	X	X		
Workmanship and soldering	3.1	4.5	X	X		
Performance						
Crosspoint switching	3.3.2	4.5.1	X			
Manual override capability	3.3.3.4	4.5.1	X			
Alert circuit indication	3.3.4	4.5.1	X			
Rotary switch function select	3.3.3.1	4.5.1	X			
Supply line variation	3.4.5	4.5.1	X	X		
Insertion loss	3.2.2.1	4.5.1	X	X		
Characteristic impedance	3.2.2.2	4.5.1	X	X		
Crosstalk isolation	3.2.2.3	4.5.1	X	X		
Temperature						
Low temperature	3.5.1	4.5	X	X		
High temperature	3.5.1	4.5	X		X	
Humidity	3.1	4.5	X			X
Workmanship screen	3.6.7	4.5.7	X	X		
Interchangeability	3.1	4.5.5	X			X
Thermal survey	3.1	4.5	X			
Shock	3.1	4.5	X			X
Vibration	3.1	4.5	X			X
Inclination	3.1	4.5	X			X
Enclosure	3.4.2, 3.6.1.2	4.5	X		X	X
Steady-state voltage and frequency	3.4.5	4.5	X		X	
Transient voltage	3.4.5	4.5	X		X	
Transient frequency	3.4.5	4.5	X		X	
Spike voltage	3.4.5	4.5	X		X	
Power interruption	3.3.7, 3.4.5	4.5	X		X	
Power and power factor	3.4.5	4.5	X		X	
Leakage current	3.1	4.5.6	X		X	
EMI	3.6.2	4.5.4	X			X
Temperature cycling	3.6.7	4.5.7.2	X	X		
Reliability test and demonstration	3.4.6	4.5.2	X			
Maintainablity demonstration	3.4.7	4.5.3	X			

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4.4.2.1 Sampling plan for Group B inspection. Switchboards submitted for Group B inspection shall be selected by the Government Quality Assurance Representative and shall be representative of current production. Group B inspection, including sampling, shall conform to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall be performed on inspection lots that have passed Group A inspection. The acceptable quality level shall be 6.5 percent defective and the inspection level shall be S-3 for normal, tightened, and reduced inspection.

4.4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective switchboards and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be conducted on a sampling basis as specified in 4.4.3.1. The inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the equipment as a whole; simulated service tests; tests of the effects of changes of environment (such as extremes of temperature and humidity); and tests of the effects of shock, vibration, and inclination. Environmental inspection shall include the examinations and tests specified in TABLE III (Group C).

4.4.3.1 Sampling for environmental inspection. One switchboard shall be selected from each successive 50 switchboards, or fraction thereof, produced. The first sample switchboard shall be selected from the first month's production.

4.4.3.2 Nonconforming environmental sample switchboards. If a sample switchboard fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall implement the corrective action necessary to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. The switchboards shall then be retested.

4.4.4 Reinspection of production control and environmental test switchboards. Unless otherwise specified in the contract (see 6.2.1), sample switchboards which have been subjected to, and have passed, production control and environmental inspections may be accepted on the contract provided they are resubjected to, and pass, the production inspection specified in 4.4.1 after repair of all visible damage.

4.5 Test methods. Unless otherwise specified in 4.5.1 through 4.5.7.2, the examinations and tests listed in TABLE III shall be performed in accordance with the test methods specified in MIL-E-16400.

4.5.1 Performance tests. The switchboards shall be subjected to a performance test including all operating controls to determine compliance with the safety requirements specified in 3.1 and the performance parameters specified in 3.2 and 3.3. The performance tests shall include, as a minimum, the individual characteristics specified in TABLE III.

4.5.2 Reliability testing. Reliability testing shall be performed in accordance with MIL-STD-781, Test Plan IVC. Test temperature shall be maintained at 50° Celsius (C). Vibration stress (continuous) shall be as specified in the Combined environments for shipboard equipment paragraph of MIL-STD-781. Test conditions and parameters to be measured during test shall be as follows:

a. The electrical switching load shall consist of the following:

1. 600-ohm balanced audio (20 Hz to 5 kHz)

2. ±28 VDC (controls)

3. 600-ohm unbalanced audio (20 Hz to 20 kHz)

4. 115 VAC, 60 Hz

b. The frequency of switching shall be 100,000 cycles with 30 seconds per position.

c. One cycle shall be designated as the sequence of stepping through all positions of the switch.

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4.5.3 Maintainability demonstration.

4.5.3.1 Maintainability equipment demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be performed by qualified technicians and shall be used to verify that the equipment corrective maintenance requirements are met.

4.5.3.2 Demonstrated conditions. The equipment maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual PC boards) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. Sixty candidate corrective maintenance faults shall be determined in accordance with Appendix A of MIL-STD-471. The following information shall be made available for each candidate task:

a. Designation of specific faulty part

b. Failure mode

c. Means of introducing fault (substitution of faulty part of simulation thereof)

The procuring activity or its authorized representative shall use the candidate tasks as a guide to select a sample of 20 tasks for the demonstration.

4.5.3.3 Subassembly (module) maintainability. The module maintainability requirements shall be demonstrated by replacement of piece parts at the organizational, intermediate, or depot-level, as appropriate. Fault selection and simulation for each repairable module shall be in accordance with the procedure for the equipment maintainability demonstration. A total of 20 faults shall be selected by the procuring activity for the module maintainability demonstration.

4.5.3.4 Accept or reject criteria.

4.5.3.4.1 Corrective maintenance. The accept or reject criteria for the demonstration of the corrective maintenance times for the demonstration shall be as specified in TABLE IV.

TABLE IV. Acceptance criteria.

	M_{ct}	M_{maxct}
Acceptance level	8	0
Sample size	20	20

Acceptance shall occur when the number of observed corrective maintenance task times which exceed the required value of each specified index (M_{ct}, M_{maxct}) is less than or equal to that shown in TABLE IV corresponding to each index for the specified confidence level. The duration of each task shall be compared to the required value(s) and recorded as greater than or lesser than each index. The actual value shall also be recorded. An accept decision can only be made when an accept decision is made for both M_{ct} and M_{maxct}. Accept or reject criteria for preventive maintenance shall be the capability to perform each preventive maintenance task with no degradation of system performance. If degradation of system operations exists, the preventive maintenance test will have resulted in a reject decision.

4.5.3.5 Preventive maintenance. The demonstration shall qualitatively demonstrate that on-line preventive maintenance for the switchboards shall not interfere with on-line operation.

4.5.3.6 Demonstration environment. The demonstration at the organizational level shall be performed in an environment similar to that in which the switchboard will be installed.

4.5.3.7 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the maintainability demonstration.

4.5.3.8 Technical documentation. Technical documentation shall be limited to the technical manual and related maintenance documentation delivered with the switchboard.

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4.5.3.9 Rejection. Failure (see 6.4.1) to conform to any of the maintainability requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the procuring activity shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies

b. Upon approval of a documented approach, the contractor shall repeat the demonstration until an accept decision is reached.

4.5.3.10 Acceptance. The maintainability demonstration shall exhibit that the switchboards conform to the accept criteria prior to delivery.

4.5.4 EMI test. The requirements of 3.6.2 shall be verified by tests performed in accordance with MIL-STD-462. For Group C inspection, only tests CE03 and RS03 shall be performed.

4.5.5 Interchangeability inspection. The requirements for interchangeability specified in 3.1 shall be demonstrated on two switchboards chosen at random from each lot of 50 or less.

4.5.6 Leakage current test.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE SWITCHBOARD DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.6.1 Switchboard connections. Each switchboard directly connected to an external power source and units deriving power from the switchboard shall be placed on an insulated surface. All safety ground conductors between the switchboard and units deriving power from the switchboard shall be intact. The safety ground conductor between the switchboard under test and the source power shall be opened during the test. The switchboard shall be connected as shown in FIGURES 2 and 3 if it is connected to single-phase or DC source power. When DC measurements are performed do not reverse polarity unless the switchboard has reverse polarity protection.

4.5.6.2 Measurement. Leakage current shall be measured on the switchboard in its normal operating configuration and under maximum radio frequency loading conditions. Controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the switchboard is designed. The overall measurement error shall not exceed five percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in FIGURES 2 and 3. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.7 Workmanship screen. Prior to conducting the temperature cycling test, vibration shall be performed on each switchboard. The vibration shall be performed at the module or drawer level. All hardware, including cables and connectors, shall be exposed to vibration.

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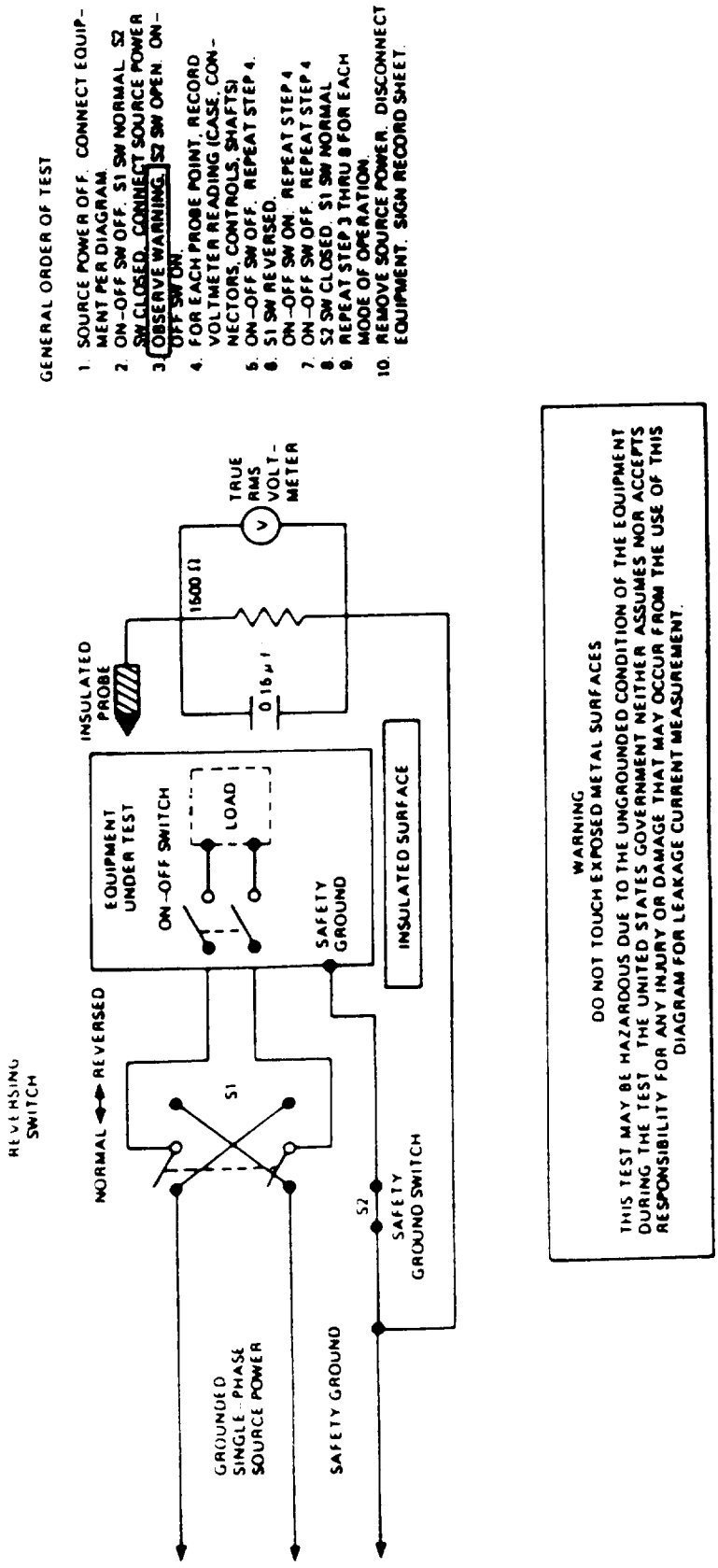


FIGURE 2. Single-phase test diagram for leakage current measurement.

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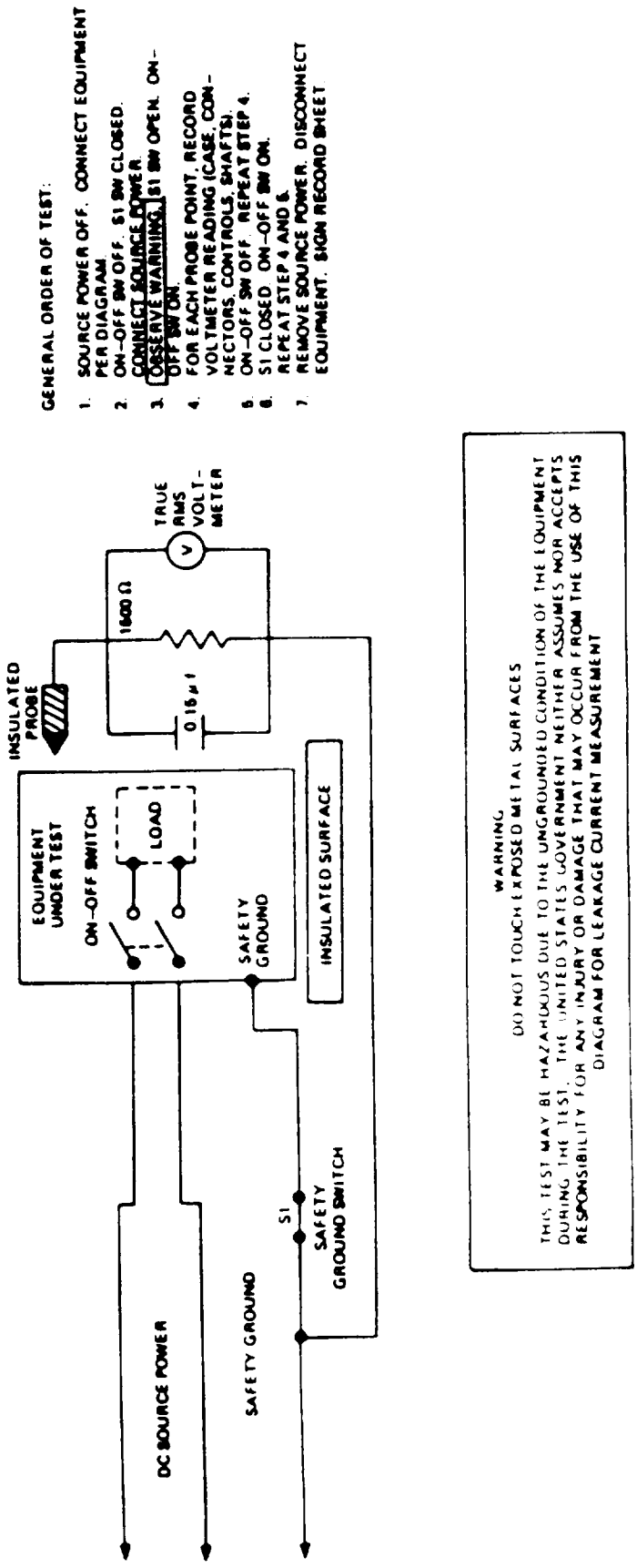


FIGURE 3. DC test diagram for leakage current measurement.

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4.5.7.1 Vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for 10 minutes in the axis deemed most susceptible to vibration excitation. All test items shall be hard-mounted (without shock isolators) and shall be subjected to the vibration conditions of FIGURE 4. The control accelerometer shall be located next to one of the mounting points of the item under test. Equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density (ASD). The test item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during test shall be corrected and the test repeated.

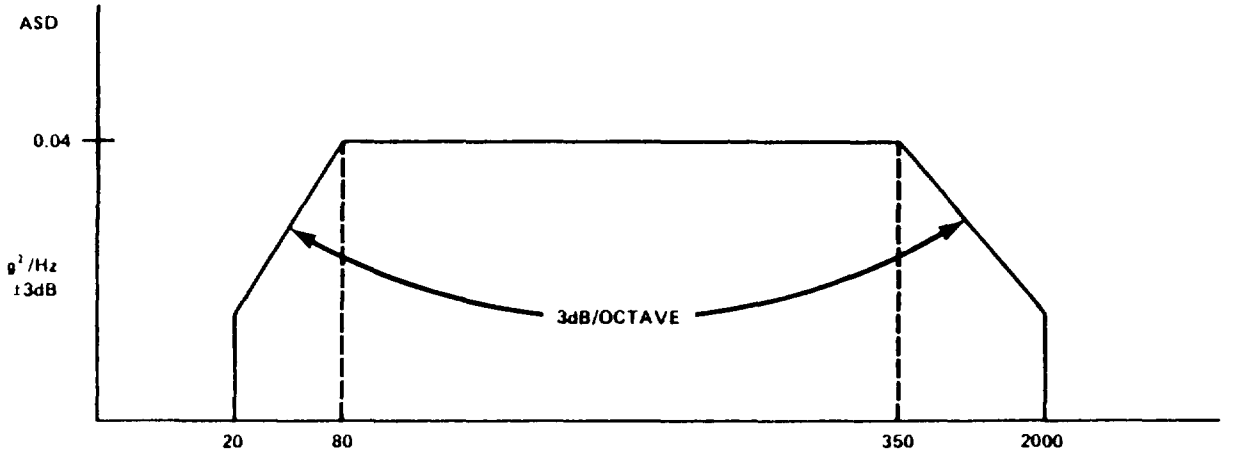


FIGURE 4. Random vibration curve.

4.5.7.2 Temperature cycling. Each switchboard shall be subjected to the temperature curve shown in FIGURE 5. The number of cycles required shall be 10 cycles. The temperature rate of change shall be not less than 5°C per minute. Switchboard power shall be turned on and off at the indicated times. When practical, the switchboard drawers, panels, and enclosures shall be opened or removed for maximum exposure to the changing temperature. The last enclosure shall be opened or removed for maximum exposure to the changing temperature. The last temperature cycle shall be failure-free. When performance measurements are required, a functional, operating test shall be performed. The dwell time shall be 80 percent of the time required for the largest electrical or electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810 for the nonoperating condition.

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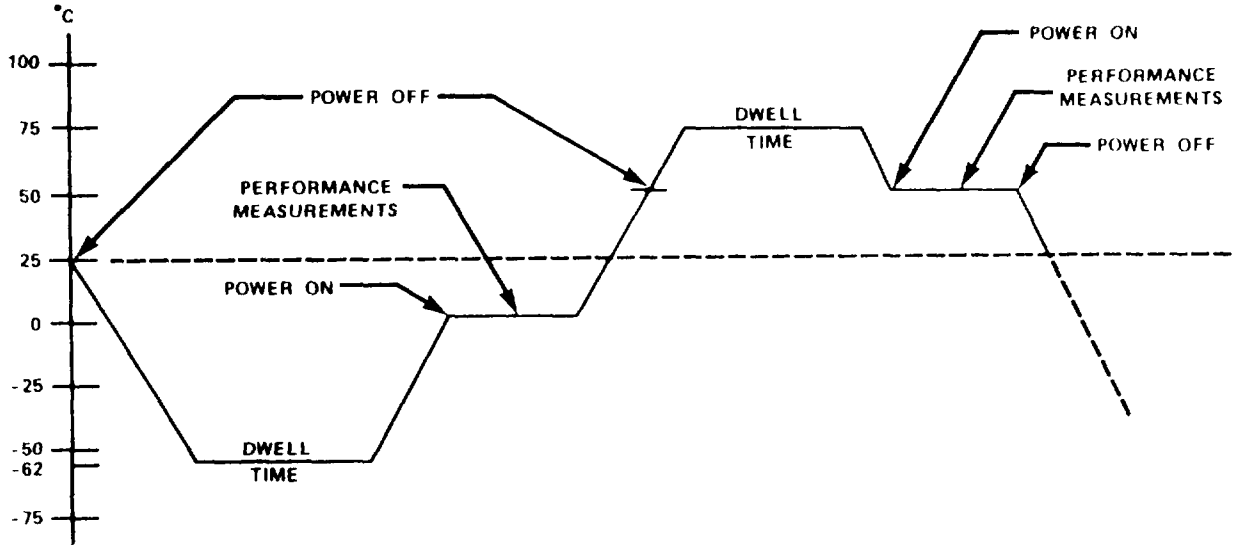


FIGURE 5. Temperature curve (one cycle).

4.6 Inspection of preparation for delivery. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The switchboard covered by this specification is intended for use in submarines.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Number of first article samples to be submitted if other than specified in 4.3

c. When reinspected production control and environmental sample units may not be accepted (see 4.4.4)

d. Levels of preservation, packaging, packing, and marking required (see 5.1)

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of Defense Acquisition Regulation (DAR) 7-1049 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below will be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
4.4	Procedures, Test	UDI-T-22710
4.4	Test Report	DI-T-5247
4.5	Report, Thermal Survey	DI-R-7036
4.5.2	Reports, Reliability Test and Demonstration	DI-R-7034
4.5.2	Procedures, Reliability Tests	DI-R-7035
4.5.3	Plan, Maintainability Demonstration	DI-R-2129
4.5.4	Report, Test, Electromagnetic Interference/Electromagnetic Compatibility (EMI/EMC)	DI-R-7062

(DIDs related to this specification, and identified in Section 6, will be approved and listed as such in DoD 5000.19L., Vol. II, AMSL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104.55 of the DAR. The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.4 Definitions. Definitions of terms used in this specification are given in 6.4.1 and 6.4.2.

6.4.1 Failure. Failure shall be any malfunction or parameter deviation that prevents the switchboard from performing within the operational requirements specified herein.

6.4.2 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service.)

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
NAVY - EC
(Project 5820-N764)

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	☐ VENDOR
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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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